AWS Cost Estimate - 50 Onboarding Events per Month

Scope and assumptions

This is a small serverless workload in `ap-south-1` (Mumbai). The estimate is intentionally conservative and intended for an academic/portfolio submission rather than billing reconciliation.

Assumptions per month:

- 50 new hires
- 3 documents per employee = 150 uploaded documents
- Average document size: 5 MB (about 0.75 GB new storage/month)
- About 1,000 HTTP API calls total
- About 1,500 Lambda invocations total
- About 1,500 Step Functions Standard state transitions total
- About 2,500 DynamoDB writes and 5,000 reads
- 50 Cognito monthly active users
- About 250 onboarding/reminder emails if SES is used
- Low-volume CloudFront traffic (well below 100 GB and 1 million requests)
- Small CloudWatch log volume

Estimate

Service	Estimated monthly usage	Approx. monthly cost
AWS Lambda	~1,500 requests, short executions	~$0.00 at this scale; well below Lambda monthly free-tier request/compute allowances in typical eligible accounts

API Gateway HTTP API	~1,000 requests	~$0.00-$0.01
Step Functions Standard	~1,500 transitions	~$0.00 because this is below the 4,000-transition monthly free tier
DynamoDB On-Demand	~2,500 writes + 5,000 reads, tiny storage	<$0.01-$0.05
Amazon S3	~0.75 GB added/month + PUT/GET requests	~$0.02-$0.10
Amazon Cognito	50 direct-sign-in MAUs	$0.00 for Lite/Essentials because the current free tier includes 10,000 MAUs/month
Amazon SES	~250 outbound emails	roughly $0.03-$0.05 depending on pricing option, excluding attachment/data charges
Amazon SNS	~50 HR email notifications	negligible for this workload
Amazon CloudFront	low-volume portfolio traffic	~$0.00 under the current free plan allowance if eligible/selected
Amazon CloudWatch	small Lambda/Step Functions logs	~$0.00-$0.50 depending on log volume/retention

Estimated total

Expected monthly cost: approximately $0-$2 per month for 50 onboardings under light usage, with a conservative budget ceiling of about $5/month.

The dominant variables are CloudWatch log ingestion/retention, S3 document retention, data transfer, and whether the account is using applicable AWS free-tier/free-plan allowances.

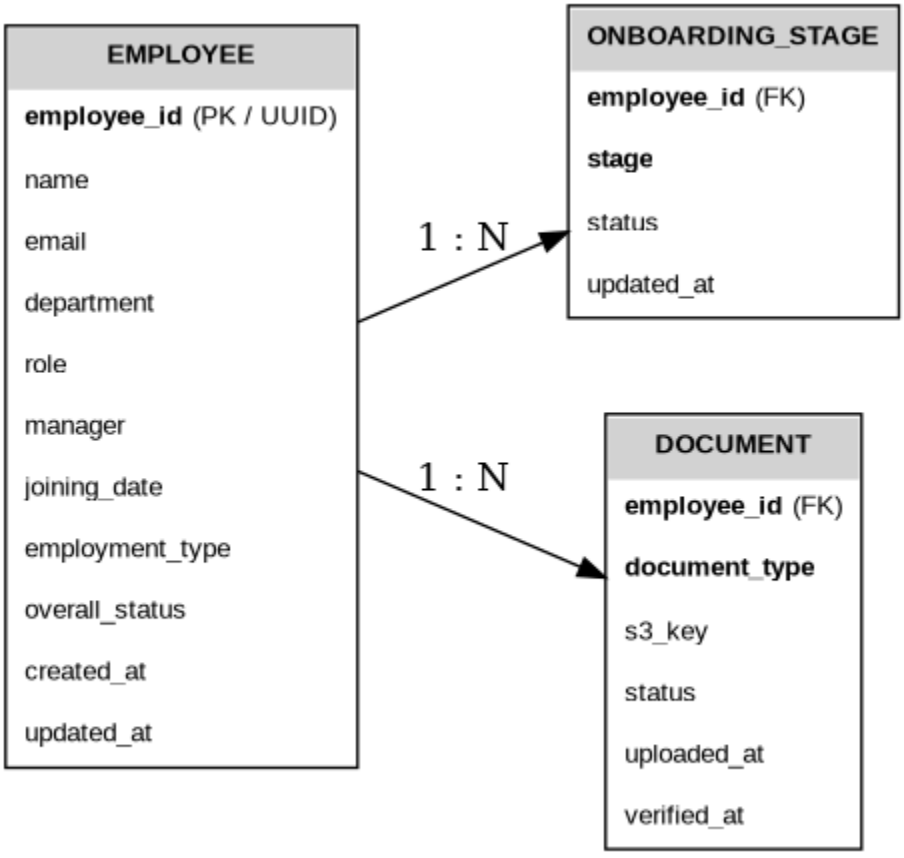
Important note

AWS pricing changes over time and varies by Region. The AWS Pricing Calculator should be used for a production procurement estimate. This document is an academic estimate for the requested 50 onboarding events/month.

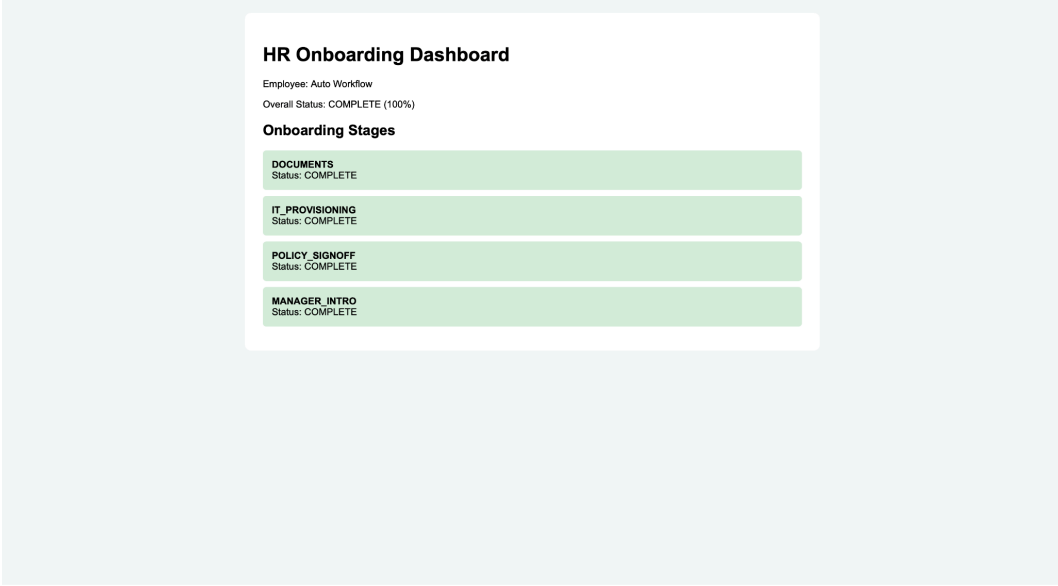
References

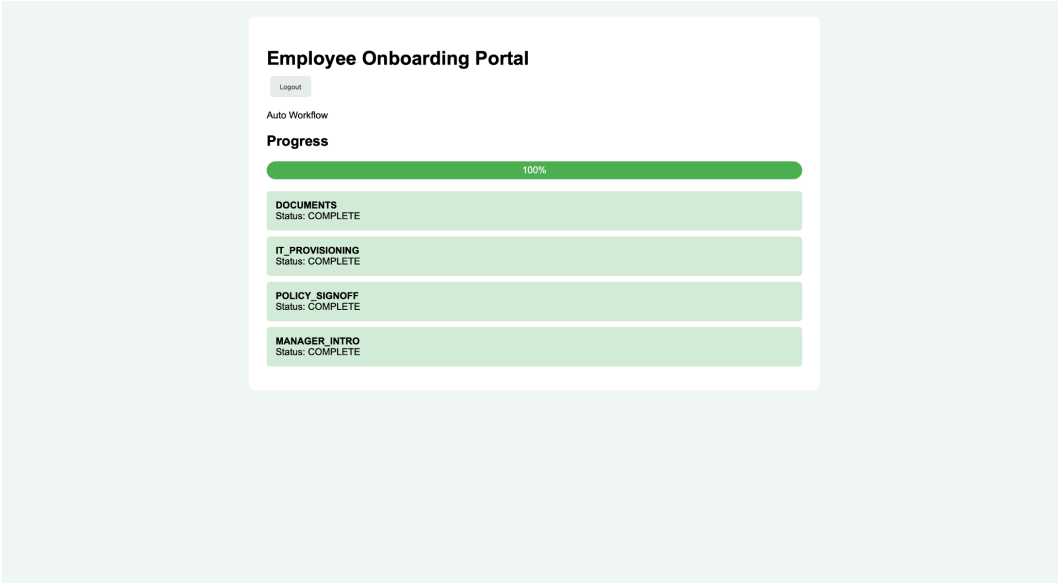
- AWS Lambda Pricing - free tier includes 1 million requests and 400,000 GB-seconds/month; standard request pricing is $0.20 per million requests. https://aws.amazon.com/lambda/pricing/
- AWS Step Functions Pricing - Standard Workflows are billed per state transition; 4,000 state transitions/month are included in the ongoing free tier. https://aws.amazon.com/step-functions/pricing/
- Amazon API Gateway Pricing - HTTP APIs are pay-per-request, with pricing examples around $1.00 per million requests at the first tier in US East; regional prices vary. https://aws.amazon.com/api-gateway/pricing/
- Amazon DynamoDB Pricing - on-demand mode bills per read/write request unit and storage. https://aws.amazon.com/dynamodb/pricing/
- Amazon Cognito Pricing - Lite and Essentials currently include 10,000 direct/social MAUs per month in the ongoing free tier. https://aws.amazon.com/cognito/pricing/
- Amazon SES Pricing - current pricing includes low per-1,000-email outbound rates; exact plan choice affects price. https://aws.amazon.com/ses/pricing/
- Amazon CloudFront Developer Guide - current Free plan allowance includes 1 million requests and 100 GB data transfer/month. https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/flat-rate-pricing-plan.html

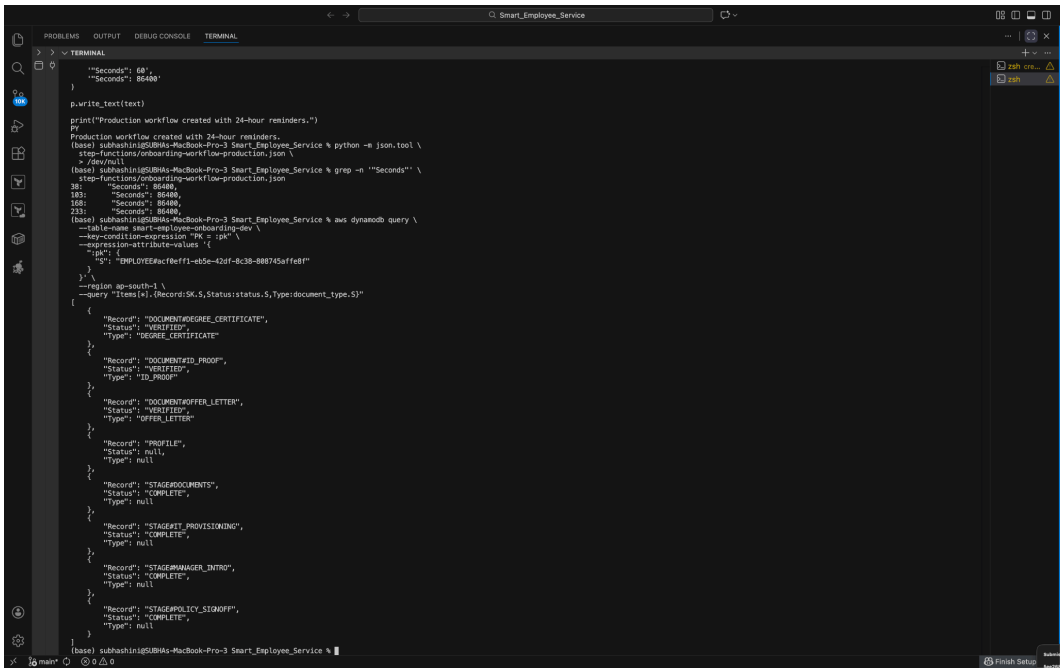
ER Diagram (DynamoDB Employee Schema)

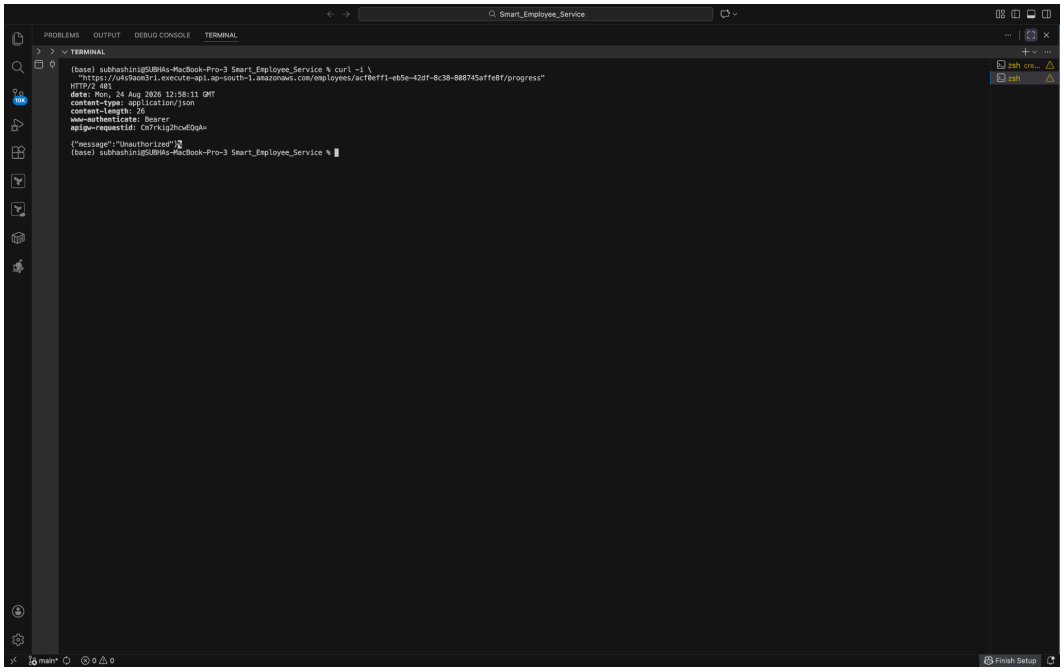


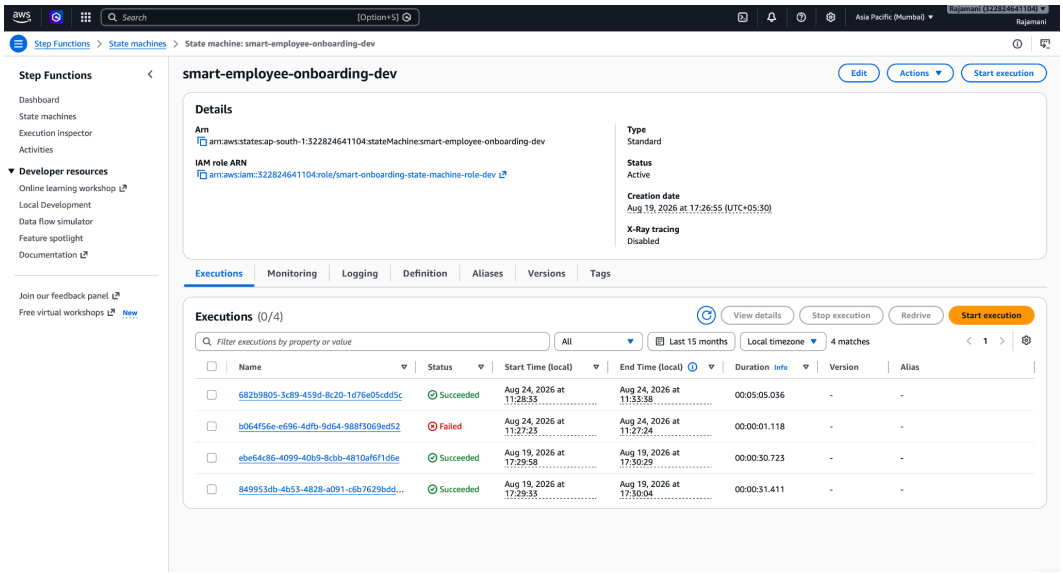
Screenshots

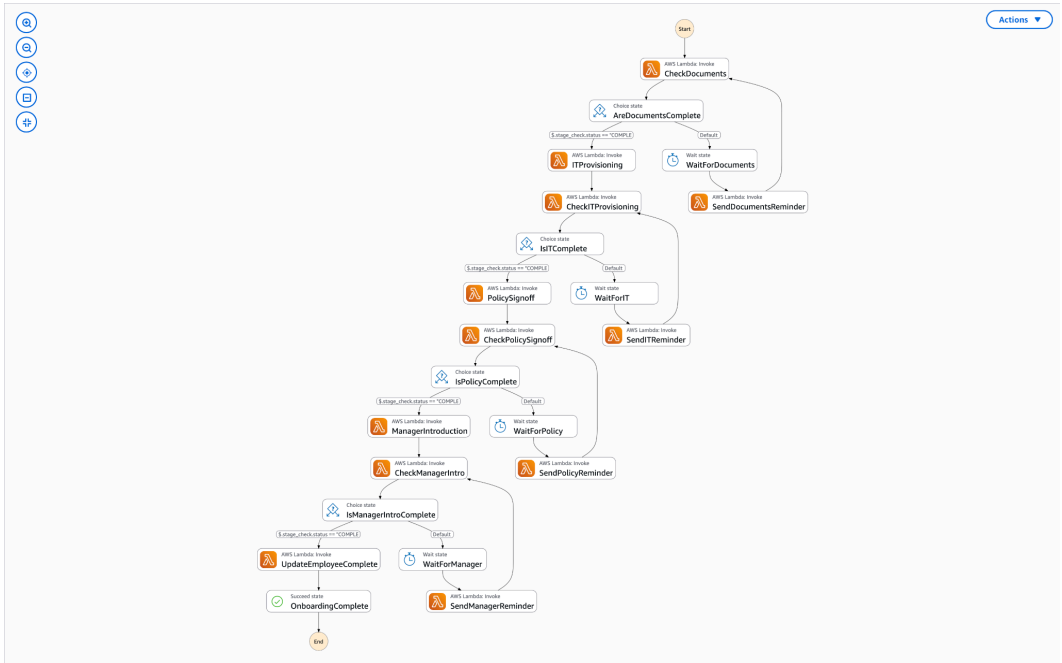


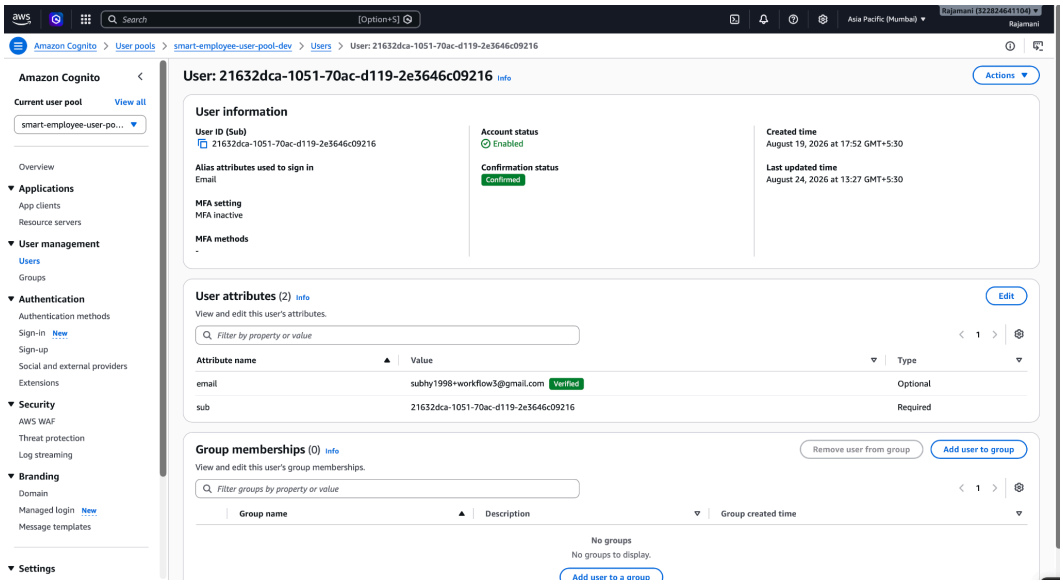



Smart_Employee_Service
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
TERMINAL
zsh cre...
zsh
"Seconds": 60',
"Seconds": 86400'
)
p.write_text(text)
print("Production workflow created with 24-hour reminders.")
PY
Production workflow created with 24-hour reminders.
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service % python -m json.tool \
step-functions/onboarding-workflow-production.json \
> /dev/null
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service % grep -n '"Seconds"' \
step-functions/onboarding-workflow-production.json
38: "Seconds": 86400,
103: "Seconds": 86400,
168: "Seconds": 86400,
233: "Seconds": 86400,
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service % aws dynamodb query \
--table-name smart-employee-onboarding-dev \
--key-condition-expression "PK = :pk" \
--expression-attribute-values '{
":pk": {
"S": "EMPLOYEE#acf0eff1-eb5e-42df-8c38-808745affe8f"
}
}' \
--region ap-south-1 \
--query "Items[*].{Record:SK.S,Status:status.S,Type:document_type.S}"
[
{
"Record": "DOCUMENT#DEGREE_CERTIFICATE",
"Status": "VERIFIED",
"Type": "DEGREE_CERTIFICATE"
},
{
"Record": "DOCUMENT#ID_PROOF",
"Status": "VERIFIED",
"Type": "ID_PROOF"
},
{
"Record": "DOCUMENT#OFFER_LETTER",
"Status": "VERIFIED",
"Type": "OFFER_LETTER"
},
{
"Record": "PROFILE",
"Status": null,
"Type": null
},
{
"Record": "STAGE#DOCUMENTS",
"Status": "COMPLETE",
"Type": null
},
{
"Record": "STAGE#IT_PROVISIONING",
"Status": "COMPLETE",
"Type": null
},
{
"Record": "STAGE#MANAGER_INTRO",
"Status": "COMPLETE",
"Type": null
},
{
"Record": "STAGE#POLICY_SIGNOFF",
"Status": "COMPLETE",
"Type": null
}
]
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service %
main*
0 0
Finish Setup


Smart_Employee_Service
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
TERMINAL
zsh cre...
zsh
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service % curl -i \
"https://u4s9aok3rl.execute-api.ap-south-1.amazonaws.com/employees/acf0eff1-eb5e-42df-8c38-808745affe8f/progress"
HTTP/2 401
date: Mon, 24 Aug 2026 12:58:11 GMT
content-type: application/json
content-length: 26
www-authenticate: Bearer
apigw-requestid: Cn7rkig2hcwEQqA=
{"message":"Unauthorized"}
(base) subhashini@SUBHAs-MacBook-Pro-3 Smart_Employee_Service %
main*
0 0
Finish Setup


Search
[Option+S]
Asia Pacific (Mumbai)
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Step Functions
State machines
State machine: smart-employee-onboarding-dev
Step Functions
Dashboard
State machines
Execution inspector
Activities
Developer resources
Online learning workshop
Local Development
Data flow simulator
Feature spotlight
Documentation
Join our feedback panel
Free virtual workshops
New
smart-employee-onboarding-dev
Edit
Actions
Start execution
Details
Arn
arn:aws:states:ap-south-1:322824641104:stateMachine:smart-employee-onboarding-dev
IAM role ARN
arn:aws:iam::322824641104:role/smart-onboarding-state-machine-role-dev
Type
Standard
Status
Active
Creation date
Aug 19, 2026 at 17:26:55 (UTC+05:30)
X-Ray tracing
Disabled
Executions
Monitoring
Logging
Definition
Aliases
Versions
Tags
Executions (0/4)
View details
Stop execution
Redrive
Start execution
Filter executions by property or value
All
Last 15 months
Local timezone
4 matches
1
Name
Status
Start Time (local)
End Time (local)
Duration
Info
Version
Alias
682b9805-3c89-459d-8c20-1d76e05cdd5c
Succeeded
Aug 24, 2026 at 11:28:33
Aug 24, 2026 at 11:33:38
00:05:05.036
-
-
b064f56e-e696-4dfb-9d64-988f3069ed52
Failed
Aug 24, 2026 at 11:27:23
Aug 24, 2026 at 11:27:24
00:00:01.118
-
-
ebe64c86-4099-40b9-8cbb-4810af6f1d6e
Succeeded
Aug 19, 2026 at 17:29:58
Aug 19, 2026 at 17:30:29
00:00:30.723
-
-
849953db-4b53-4828-a091-c6b7629bdd...
Succeeded
Aug 19, 2026 at 17:29:33
Aug 19, 2026 at 17:30:04
00:00:31.411
-
-


Actions
Start
AWS Lambda: Invoke
CheckDocuments
Choice state
AreDocumentsComplete
$.stage_check.status == "COMPLE"
Default
AWS Lambda: Invoke
ITProvisioning
Wait state
WaitForDocuments
AWS Lambda: Invoke
CheckITProvisioning
AWS Lambda: Invoke
SendDocumentsReminder
Choice state
IsITComplete
$.stage_check.status == "COMPLE"
Default
AWS Lambda: Invoke
PolicySignoff
Wait state
WaitForIT
AWS Lambda: Invoke
CheckPolicySignoff
AWS Lambda: Invoke
SendITReminder
Choice state
IsPolicyComplete
$.stage_check.status == "COMPLE"
Default
AWS Lambda: Invoke
ManagerIntroduction
Wait state
WaitForPolicy
AWS Lambda: Invoke
CheckManagerIntro
AWS Lambda: Invoke
SendPolicyReminder
Choice state
IsManagerIntroComplete
$.stage_check.status == "COMPLE"
Default
AWS Lambda: Invoke
UpdateEmployeeComplete
Wait state
WaitForManager
Succeed state
OnboardingComplete
AWS Lambda: Invoke
SendManagerReminder
End


Search
[Option+S]
Asia Pacific (Mumbai)
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Rajamani
Amazon Cognito
User pools
smart-employee-user-pool-dev
Users
User: 21632dca-1051-70ac-d119-2e3646c09216
Amazon Cognito
Current user pool
View all
smart-employee-user-po…
Overview
Applications
App clients
Resource servers
User management
Users
Groups
Authentication
Authentication methods
Sign-in New
Sign-up
Social and external providers
Extensions
Security
AWS WAF
Threat protection
Log streaming
Branding
Domain
Managed login New
Message templates
Settings
User: 21632dca-1051-70ac-d119-2e3646c09216 Info
Actions
User information
User ID (Sub)
21632dca-1051-70ac-d119-2e3646c09216
Alias attributes used to sign in
Email
MFA setting
MFA inactive
MFA methods
-
Account status
Enabled
Confirmation status
Confirmed
Created time
August 19, 2026 at 17:52 GMT+5:30
Last updated time
August 24, 2026 at 13:27 GMT+5:30
User attributes (2) Info
Edit
View and edit this user's attributes.
Filter by property or value
Attribute name
Value
Type
email
subhy1998+workflow3@gmail.com
Verified
Optional
sub
21632dca-1051-70ac-d119-2e3646c09216
Required
Group memberships (0) Info
Remove user from group
Add user to group
View and edit this user's group memberships.
Filter groups by property or value
Group name
Description
Group created time
No groups
No groups to display.
Add user to a group

3 of many

Employee documents verified

Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 11:03 AM (7 hours ago)
to subhy1998

All required onboarding documents for employee acf0eff1-eb5e-42df-8c38-808745affe8f have been received and verified.

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https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:322824641104:smart-hr-onboarding-dev:2a819d3f-1b7f-4155-abf8-9ae5d08a4ed6&Endpoint=subhy1998@gmail.com

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Employee Onboarding Reminder

Inbox ×

AWS Notifications 11:20 AM (7 hours ago)
Reminder: onboarding stage IT_PROVISIONING is still incomplete for employee acf0eff1-eb5e-42df-8c38-808745affe8f. -- If you wish to stop receiving notifications

AWS Notifications <no-reply@sns.amazonaws.com> 11:29 AM (7 hours ago)
to subhy1998

Reminder: onboarding stage IT_PROVISIONING is still incomplete for employee acf0eff1-eb5e-42df-8c38-808745affe8f.

--
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AWS Notifications 11:30 AM (7 hours ago)
Reminder: onboarding stage IT_PROVISIONING is still incomplete for employee acf0eff1-eb5e-42df-8c38-808745affe8f. -- If you wish to stop receiving notifications

AWS Notifications <no-reply@sns.amazonaws.com> 11:31 AM (7 hours ago)
to subhy1998

AWS Notifications <no-reply@sns.amazonaws.com> 11:32 AM (7 hours ago)
to subhy1998

Reminder: onboarding stage POLICY_SIGNOFF is still incomplete for employee acf0eff1-eb5e-42df-8c38-808745affe8f.

AWS Notifications <no-reply@sns.amazonaws.com> 11:33 AM (7 hours ago)
to subhy1998

Reminder: onboarding stage MANAGER_INTRO is still incomplete for employee acf0eff1-eb5e-42df-8c38-808745affe8f.

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